HYPERTENSION ALERT SYSTEM FOR PREGNANCY

Hack-A-Roo 2026 | Pitch-to-Build Track

CS 5588 - Data Science Capstone

University of Missouri-Kansas City

Team project report prepared by

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Research prototype using simulated patients - not for clinical use.

1. Executive Summary

Severe hypertension during pregnancy and the postpartum period requires timely recognition, repeat blood pressure measurements, communication, and appropriate treatment. In the original clinical pitch, the team reported that 69% of persistent severe-hypertension events on their labor and delivery unit received antihypertensive treatment within the recommended 60-minute window. In response to this challenge, our interdisciplinary team developed the Hypertension Alert System for Pregnancy to help healthcare staff track time-sensitive clinical tasks and reduce reliance on memory alone.

The prototype combines configurable, rule-based episode tracking with unit and bedside dashboards, reminders, escalation, and draft electronic medical record (EMR) notes. We also developed a synthetic patient workflow simulator to evaluate the timing of clinical activities and explore how changes in workflow delays could affect treatment. The application is not connected to a live clinical EHR and does not provide treatment recommendations or administer medication.

Using 80 synthetic patients, we evaluated a scenario in which selected workflow delays were reduced by an assumed 50%. Among 20 matched patient pairs, the average time to the first medication dose decreased from 66.20 minutes without the tool to 55.85 minutes with the simulated tool, a difference of 10.35 minutes. Similarly, the average repeat-reading lateness decreased from 5.70 to 2.86 minutes among 50 patients with severe readings. These findings illustrate the potential effect of shorter workflow delays under the model's assumptions, rather than demonstrating improvements in actual clinical practice.

2. Clinical Problem and Context

Persistent severe-range hypertension during pregnancy and the postpartum period requires prompt clinical attention. For this project, severe-range blood pressure was defined as a systolic reading of at least 160 mmHg or a diastolic reading of at least 110 mmHg. When a severe reading occurs, the clinical workflow involves repeating the measurement to determine whether the elevation persists, notifying the appropriate healthcare provider, and initiating treatment when indicated. The recommended treatment window is 60 minutes from the first severe reading.

The clinical pitch team identified four points where delays could occur: recognizing the initial severe reading, obtaining a repeat measurement, contacting the appropriate provider, and initiating treatment. These activities take place in a busy clinical environment where nurses and providers are responsible for multiple patients and competing tasks. Our system was designed to support this workflow by making episode status, elapsed time, and pending actions visible to healthcare staff.

The project focuses on supporting timely clinical responses and improving workflow coordination while leaving treatment decisions to healthcare professionals. Pregnant and postpartum patients, their infants, and healthcare providers could potentially benefit from this approach. However, the prototype has not been evaluated in a clinical setting, and further testing and validation would be necessary to establish its safety, effectiveness, and impact on patient outcomes.

3. Project Objectives and Requirements

Our goal was to develop a working prototype that supports the timely recognition, monitoring, and management of severe-hypertension episodes during pregnancy and the postpartum period. The system was designed to use explicit, rule-based logic for clinical alerts, with artificial intelligence (AI) limited to supporting functions such as documentation.

The project had six main objectives:

1. **Identify and track severe-hypertension episodes:** Apply predefined blood pressure thresholds, record the initial severe reading, and track the 60-minute treatment window.
2. **Support timely reassessment:** Monitor when repeat blood pressure measurements are due and identify overdue reassessments.
3. **Improve communication and escalation:** Track episode confirmation, provider acknowledgment, and escalation events to support timely clinical responses.
4. **Provide clear patient monitoring interfaces:** Develop unit-level and bedside dashboards that display patient status, blood pressure readings, and relevant workflow information.
5. **Support accurate clinical documentation:** Generate draft electronic medical record (EMR) notes using recorded episode events, with safeguards to ensure that documented facts can be traced to the underlying records.

6. **Evaluate system behavior using synthetic data:** Compare baseline and simulated alert-supported workflows, assess differences in reassessment and treatment timing, and examine how the findings change under different assumptions.

4. Design Process and Human-AI Co-Design

Our development process followed five stages: Human Design, AI Design, Human–AI Co-Design, Implementation, and Evaluation. We began by translating the clinical pitch team's workflow requirements into a system design that addressed severe blood pressure recognition, reassessment, provider communication, and escalation.

AI-assisted development supported parts of the coding, testing, and documentation process. As part of our Human–AI Co-Design approach, we reviewed and refined proposed solutions through team discussions, code reviews, and testing. Human judgment guided the final design, particularly our decision to use predefined blood pressure thresholds and timing rules for clinical alerts rather than relying on AI-generated decisions.

During implementation, we integrated the episode tracker with the synthetic patient generator, workflow simulation, and application, incorporating feedback on clinical timing, escalation behavior, and simulation assumptions. We also kept optional AI-generated EMR note summaries separate from clinical alert decisions.

Finally, we evaluated the system using automated tests, matched synthetic patient simulations, and sensitivity analysis to examine how assumed reductions in workflow delays affected reassessment and treatment timing. This evaluation assessed the prototype's technical behavior rather than its effectiveness in clinical practice.

5. System Architecture and Implementation

5.1 Shared Rule Engine and Protocol

The system uses a shared protocol configuration that defines severe blood pressure thresholds and the timing requirements for each episode. The EpisodeTracker processes recorded blood pressure measurements and workflow events to monitor changes in episode status. An initial severe reading opens an episode and starts the 60-minute treatment clock, with a repeat measurement scheduled for 15 minutes later. Persistent severe hypertension can be identified through subsequent severe readings or the protocol's missed-reassessment logic. The system also tracks provider notification and acknowledgment and applies escalation rules when an acknowledgment is overdue. Escalation occurs at the earlier of the configured acknowledgment timeout or 10 minutes before the treatment deadline. These timing settings are part of the prototype's configuration rather than independently validated clinical requirements. Recurrent severe readings within the same hour can also be tracked as part of an existing episode.

5.2 Application Interfaces

The prototype was developed using Streamlit and provides two main interfaces: a unit-level dashboard and a bedside view. The unit dashboard displays multiple simulated patients and their current episode statuses, allowing users to identify cases requiring attention. The bedside view provides more detailed information about an individual patient, including blood pressure readings, episode timelines, upcoming deadlines, and recorded staff actions. The demonstration includes an eight-bed unit with different hypertension scenarios. The application uses simulated patient data and is not connected to live bedside monitors or electronic health records.

5.3 Traceable EMR Documentation

The documentation component generates draft electronic medical record (EMR) notes from recorded episode events. A rule-based template creates the clinical timeline, while an optional large language model, such as Claude or a locally configured model through Ollama, can generate a narrative summary. To maintain traceability, factual details such as blood pressure measurements, timestamps, and patient identifiers must be supported by recorded events or protocol values. If an AI-generated summary fails these checks, the system uses its configured fallback rather than accepting unsupported information. The resulting note is intended for nurse review and is not automatically written to a clinical EMR.

5.4 Integration of the Application and Workflow Simulation

The application and workflow simulation share the same episode-tracking logic but serve different purposes. The application displays patient information and tracks clinical workflow events, while the simulation generates synthetic blood pressure patterns, models reassessment and staff responses, and calculates evaluation

measures. Using the same EpisodeTracker helps maintain consistency between the system's behavior and its evaluation. However, the simulated workflow includes assumptions about staff response times and treatment decisions that have not been validated in a clinical environment.

6. Synthetic Data and Workflow Model

We generated 80 synthetic patients across eight clinical scenario groups, with ten patients in each group. Of these, 50 had at least one severe blood pressure reading, while 30 did not. The scenarios represented non-hypertensive, mild-hypertension, borderline-severe, persistent-severe, nonpersistent-severe, recurrent-severe, delayed or missing reassessment, and spurious-severe patterns. All patient data used in the evaluation were synthetic, with no protected health information involved.

The workflow simulation distinguishes between a patient's underlying simulated blood pressure and the measurements actually recorded during reassessment. It generates blood pressure patterns, schedules repeat measurements and staff actions, and uses the shared EpisodeTracker to process recorded readings and monitor episode progression. The model also simulates provider responses and treatment decisions, allowing us to examine how changes in workflow timing affect the evaluated outcomes.

The repository additionally includes a loader and benchmark scenario for the UCI Maternal Health Risk dataset, which contains 1,014 records. However, this external dataset was not used to generate the patient measurements in the application or the coupled workflow simulation.

7. Evaluation Methods

7.1 Paired Baseline-versus-Tool Simulation

We evaluated the same synthetic patients under two conditions: a baseline workflow without alert support and a simulated alert-supported workflow. In the alert-supported condition, selected delays in recognizing severe readings, performing reassessments, and contacting providers were reduced by an assumed percentage. These improvements were modeling assumptions rather than measurements from clinical practice. The first-dose timing analysis included 20 matched patient pairs with a recorded medication event in both conditions.

7.2 Outcome Definitions and Denominators

First-dose time was measured from the initial severe blood pressure reading to the first simulated medication dose. Repeat-reading lateness represented the time beyond the scheduled reassessment deadline and was evaluated across the 50 patients with severe readings.

We also assessed timely management of persistent severe-hypertension episodes within the 60-minute window. This outcome included episodes with either a simulated medication dose or documented resolution. Because differences in reassessment timing could affect whether an episode was classified as persistent, the number of qualifying episodes differed between the two conditions. Consequently, the timely-management percentages should not be interpreted as directly comparable clinical treatment effects.

7.3 Sensitivity Analysis and Testing

We performed sensitivity analysis by varying the assumed reduction in selected workflow delays from 0% to 100%. This allowed us to examine how changes in these assumptions influenced reassessment and treatment timing. The project also used automated tests to check episode-tracking behavior, timing rules, and integration across system components. The final project poster reported 616 passing automated tests.

8. Results

Table 1 summarizes the results of the baseline and simulated alert-supported workflows under an assumed 50% reduction in selected workflow delays. All findings are based on synthetic patient simulations rather than clinical observations.

Table 1. Comparison of simulated workflow outcomes

Measure	Without tool	With tool	Difference or interpretation
Mean first-dose time (20 matched pairs)	66.20 min	55.85 min	10.35 min earlier
Mean repeat-reading lateness (50 patients)	5.70 min	2.86 min	2.84 min less late
Timely management of	17/42	23/41	Denominators differ

persistent episodes			
Timely episodes with a simulated medication dose	5	12	Included in timely total
Timely episodes with documented resolution	12	11	Included in timely total

Under the assumed 50% reduction in selected delays, the simulation showed an average improvement of 10.35 minutes in first-dose timing among 20 matched patient pairs. Mean repeat-reading lateness also decreased by 2.84 minutes among the 50 patients with severe readings. Timely management included both simulated medication administration and documented resolution, with different numbers of qualifying persistent episodes between the two conditions.

The sensitivity analysis showed progressively larger modeled time savings as the assumed reduction in workflow delays increased. At a hypothetical 100% reduction, the average first-dose time saving reached 20.3 minutes, compared with 10.35 minutes at 50%. The corresponding reductions in repeat-reading lateness were 5.70 and 2.84 minutes, respectively. These findings demonstrate how the modeled workflow responds to different assumptions, not improvements established in clinical practice.

9. Limitations and Future Work

Assumed workflow improvements. The simulation assumes that selected workflow delays are reduced by a specified percentage rather than measuring improvements achieved through actual system use. Provider contact is also represented as a shorter response time without modeling the steps involved in identifying and reaching the appropriate clinician.

Idealized resolution gate. The simulation has a limitation in how it determines whether medication is still needed. In some cases, it uses a patient's underlying simulated blood pressure to cancel a medication dose even when the improvement has not been documented through a recorded measurement. This could make the simulated avoidance of unnecessary medication appear more effective than it would be in practice. A more realistic approach would base treatment decisions only on recorded blood pressure measurements and available clinical information.

Clinical scope. The synthetic cohort does not fully represent the complexity of real clinical environments, including differences in patient conditions, staffing, treatment decisions, and communication. The prototype has not been evaluated with real patients or integrated into a clinical electronic health record. Although the system tracks escalation events, the simulation does not model an additional effect of escalation on staff behavior. Therefore, the results cannot establish improvements in clinical outcomes.

Future work. The next steps would include collecting real-world workflow data with appropriate approvals, developing a more realistic model of provider availability and communication, and revising the resolution gate to rely on recorded clinical information. Further evaluation with healthcare professionals, including usability testing in a simulated clinical environment, would help assess the system's practical value. Integration with electronic health records using standards such as HL7/FHIR could also be explored after further development and validation.

10. Collaboration and Individual Contributions

10.1 Obed Akwasi Aning - Synthetic Data, Evaluation and Report Preparation

Obed developed and refined the synthetic blood pressure scenarios used to evaluate the prototype, including normal and borderline readings, persistent and recurrent severe hypertension, spurious readings, and delayed or missing reassessments. He contributed to the coupled workflow simulation by connecting generated blood pressure measurements, reassessment schedules, staff actions, and treatment events to the team's episode-tracking logic. He also developed and tested baseline-versus-tool comparisons using matched synthetic patients and sensitivity analysis to examine how assumed reductions in workflow delays affected reassessment and treatment timing. During integration, he updated tests, checked the reproducibility of results, and incorporated Daniel's feedback on clinical timing and deadline-aware escalation. Obed also prepared the shared project report, organized the methodology and evaluation findings, and reviewed the updated research poster, providing feedback on how the simulation's resolution-gate limitation was presented.

10.2 Daniel Evans - Episode Tracking, System Integration, and Project Coordination

Daniel served as the build team lead, coordinating development activities and overseeing the integration of the project's components. He developed the core episode-tracking logic and shared protocol configuration, including severe blood pressure recognition, reassessment timing, episode confirmation, provider acknowledgment, and escalation rules. His work provided the foundation used by both the application and the workflow simulation.

He also reviewed the simulation design, clinical timing assumptions, and evaluation results, providing feedback that helped refine the coupled workflow model and its tests. Daniel independently checked the cohort evaluation and sensitivity analysis, coordinated the integration of changes into the shared GitHub repository, and supported testing and documentation to maintain consistency across the system.

In addition to his technical contributions, Daniel prepared the project demonstration video and developed the updated research poster, incorporating the final application features, simulation findings, and project limitations. He also coordinated the team's final presentation materials and project deliverables.

10.3 Vince Nguyen - Application Interface and Poster Design

Vince contributed to the development of the Streamlit application, helping translate the team's episode-tracking functionality into an interactive prototype. His work focused on the user-facing interface, including the unit-level dashboard and bedside view, which display patient information, blood pressure readings, episode status, and relevant workflow events. These interfaces helped demonstrate how healthcare staff could monitor severe-hypertension episodes and follow time-sensitive clinical activities.

Vince also developed the initial research poster, organizing the project's clinical problem, system design, and implementation into a visual presentation. His contributions supported the demonstration and communication of the team's work, providing a foundation for the poster's subsequent revisions.

11. Conclusion

Our team developed a working prototype that integrates rule-based severe-hypertension episode tracking, unit and bedside dashboards, workflow reminders, escalation, and draft EMR documentation. Through synthetic patient simulations, we evaluated how assumed reductions in workflow delays could affect reassessment and treatment timing. The findings demonstrate the system's technical functionality and illustrate the potential value of better workflow coordination, but they do not establish improvements in clinical practice. Further development, testing with recorded clinical workflow data, and appropriate clinical validation would be necessary before the system could be considered for use in patient care.

12. Project Resources and Deliverables

The project source code, application, demonstration video, and research poster are available through the following links.

Resource	Location
Source code, setup instructions, and automated tests	https://github.com/devans2718/hypertension-alert-pregnancy
Streamlit demonstration application	https://hypertension-alert-pregnancy-qrzuqnona3infydlandcuc.streamlit.app/
Project demonstration video	https://www.youtube.com/watch?v=QObCJRi1yW8
Updated research poster (Pull Request #32)	https://github.com/devans2718/hypertension-alert-pregnancy/pull/32

References and Source Notes

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